



YESYOU CAN CYBERSECURE, LLC
Dallas, Texas, USA
Cybersecurity - GRC/IT Audit Training

REQUIRED PROJECT 1 (Individual)

Deadline: September 2026, 11:59 pm CST.

(30 Marks)

Please use a laptop or desktop computer for this exercise.

Q1. Scenario: You are an information security analyst tasked with ensuring the integrity of sensitive files. Your organization uses WinMD5, a hashing tool, to verify file integrity.

Task: Use WinMD5 to demonstrate the hashing process and verify the integrity of the file.

Steps:

1. Download and install WinMD5 on your computer.
2. Create or use an existing Word document and name it confidential_document.txt.
3. Use WinMD5 to generate a hash key. Confirm if the document matches
4. Modify the contents of confidential_document.txt (e.g., add a new line, change a word etc). Remember to save the changes.
5. Upload the altered confidential_document.txt into WinMD5 again.
6. Paste the hash key in the second box to check whether the display result is the same if you're using Windows or compare the hash value if you're using a MacBook. What do you observe?
7. Follow the same steps with a different Word document, but this time, do not change anything in the content. Observe the results.
8. Take screenshots of each step for both scenarios (when the document content was changed and when it was not) for comparison.

Relevant links (WinMD5)

Windows Users:

<https://www.winmd5.com/>

Mac Users:

<https://apps.apple.com/us/app/md5-master/id1244747556?mt=12>

Q2. Your organization wants to ensure the confidentiality of information exchanged between employees. The IT department has decided to implement encryption as a form of information security control, and you have been tasked with training all employees. Using the adverse software: https://services.getadverse.com/hash_crypto, demonstrate symmetric encryption to the audience. Explain the **best mechanism** that can be employed to prevent bad actors from accessing the encrypted files and the encryption key when sending them to recipients via email. **Note:** Authentication and nonrepudiation are important to the recipient.

Use two entities: Senior IT Auditor (recipient) and Staff IT Auditor (sender)

Steps:

1. Explain and include screenshots of the symmetric encryption demonstration, with appropriate comments.
2. Explain, with screenshots, the **best mechanism** that can be used to strengthen the symmetric encryption controls (Hint: Asymmetric encryption).

Q3. Your company has implemented digital signature for its activities, and you have been tasked to ensure that this new application works effectively. Using the adverse platform (<https://services.getadverse.com/adversesign>)

Follow the steps below to conduct this exercise.

- a. Upload a document and add at least two signers to the document.
- b. Configure the required fields for each signer, ensuring that the following are included: Fields signature, initials, date for each signer.
- c. Comment and include screenshots of Details (documents information & Actions), Signers, Fields. and Audit trails (include how an audit trail supports accountability, traceability, integrity, monitoring, investigation, and audit/compliance evidence.)
- d. NB: you may need partners/ classmates to serve as singers for this exercise (for reference see session 4).

Q4. As the leader of security awareness training at your company, use a website of your choice for demonstration and explain how employees can stay safe online when conducting financial transactions.

Requirements:

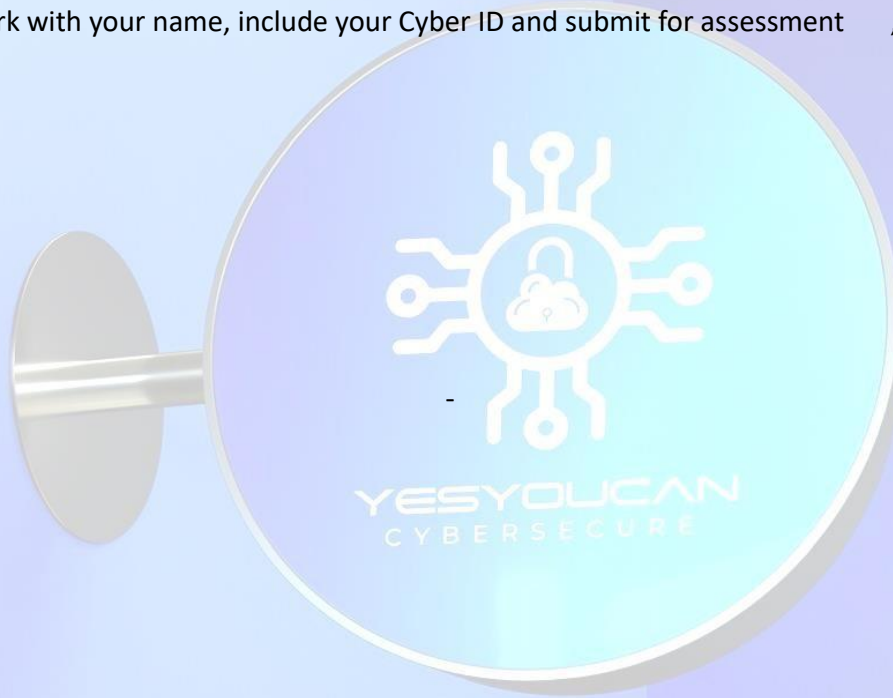
1. Take screenshots of all security measures employees should consider on the website when buying

items online.

2. Explain with screenshots, in your own words, how a web browser interacts with a website to request, retrieve, and display web content for end users as hits.

Q5. With cryptography in mind, how does WhatsApp ensure that your communication is secured?

Save the work with your name, include your Cyber ID and submit for assessment ,



NAME: CORRIE ADELAIDE BLAY

CYBER ID: 980074118

QUESTION 1

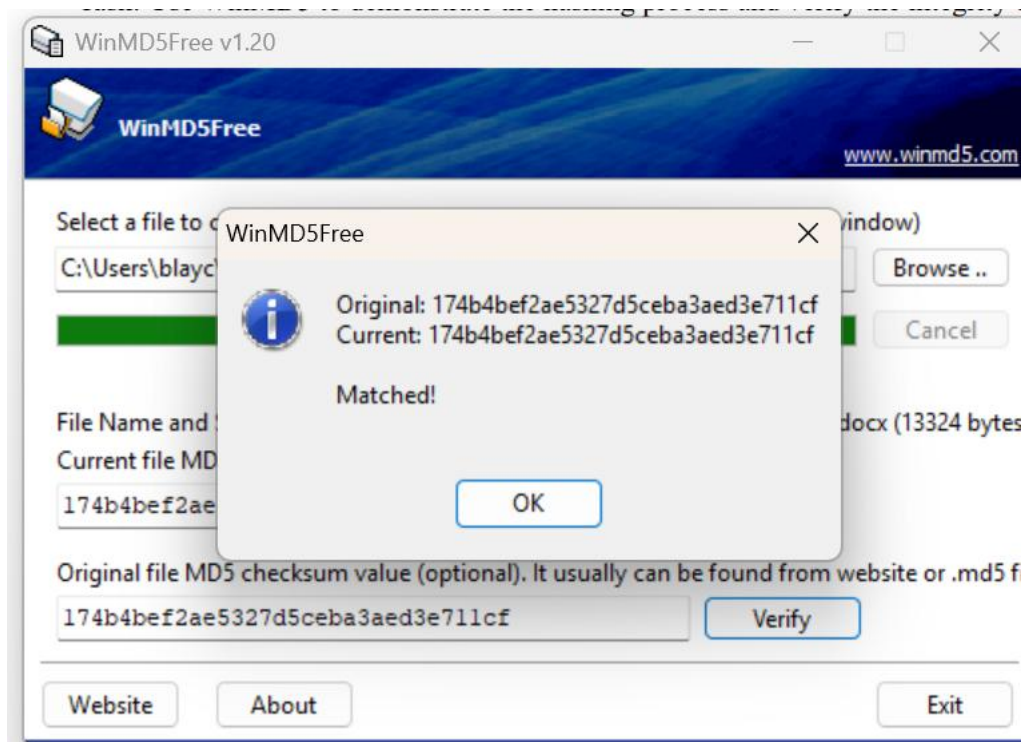
I downloaded WinMD5 from the official website and installed it on my computer.

In the first scenario, I modified the confidential_document.txt file. WinMD5 showed 'NOT Matched!' because the hash changed from **d4542142979076f67efe3a67a86871ed** to a new value, proving that even a small altercation creates a completely different hash.

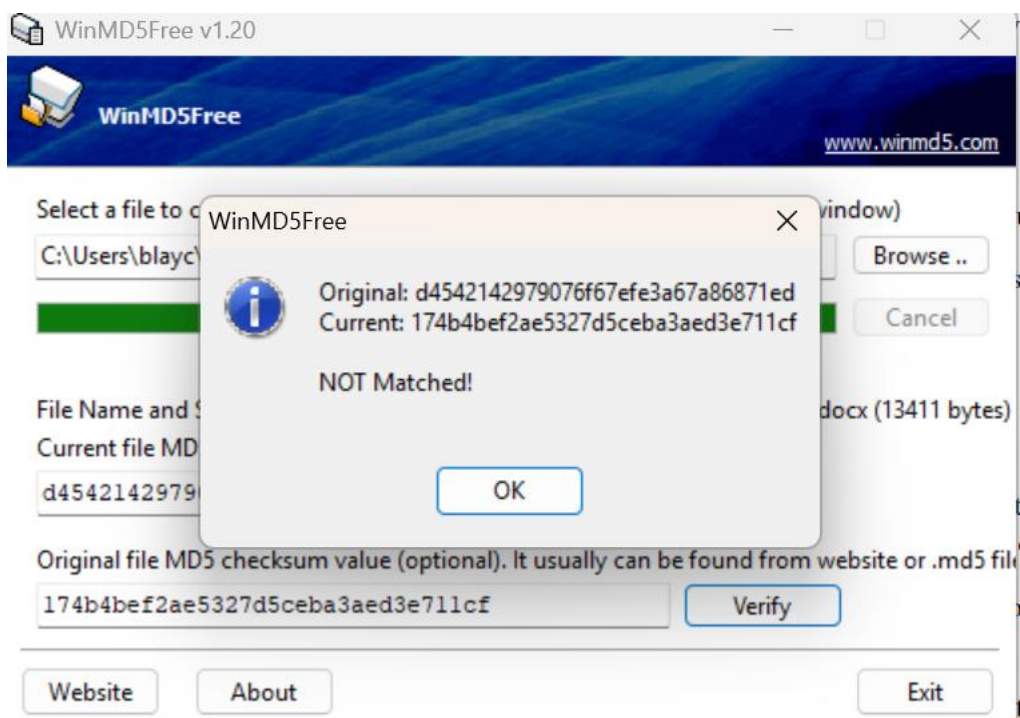
In the second scenario, I scanned unchanged_document.txt twice. WinMD5 showed 'Matched!' with the exact same hash **174b4bef2ae5327d5ceba3aed3e711cf** confirming the file was intact and unchanged.

This confirms that hash values are unique fingerprints for file content, matching hashes prove integrity, while a mismatch indicates the file has been tampered with.

Q1. Screenshot 1(Matched)



Q1. Screenshot 2 (Not Matched)



QUESTION 2 REPORT

The tool automatically detected the encryption mode from the ciphertext header and successfully decrypted the message. The warning is a standard security notice about header detection, but it did not affect the decryption process.

For Symmetric Encryption (Screenshots 1 Symmetric Encryption & Screenshot 2 Symmetric Decryption):

I used AES-256-GCM symmetric encryption with the password 'SharedKey123'. The message 'Audit findings are ready for review' was encrypted and successfully decrypted using the same password. This demonstrates symmetric encryption where one key is used for both encryption and decryption.

For Asymmetric Encryption (Screenshots 3 Asymmetric Encryption & Screenshot 4 Asymmetric Decryption):

I used RSA asymmetric encryption. A key pair was generated: a Public Key and a Private Key. The message 'This is a secure message for the Senior IT Auditor' was encrypted using the Public Key and could only be decrypted using the corresponding Private Key. This is more secure because the Private Key is never shared.

Best Mechanism to Strengthen Symmetric Encryption:

The best mechanism to prevent bad actors from accessing encrypted files is to use asymmetric encryption to securely exchange the symmetric key. The sender encrypts the symmetric

key with the recipient's Public Key. Only the recipient, who has the Private Key, can decrypt it. This ensures that even if the email is intercepted, the attacker cannot access the key or the encrypted data.

The sender should also **digitally sign** the message/file using their own private key. The recipient verifies it with the sender's public key — that's what proves authenticity and prevents the sender from denying they sent it.

Authentication is achieved because the recipient can verify that the message came from the sender using the sender's digital signature. Nonrepudiation is achieved because the message was signed with the sender's private key. Since only the sender possesses that private key, they cannot credibly deny having sent the message — the signature itself serves as proof of origin.

Screenshot 1 Symmetric Encryption:

Symmetric Encryption - Message + Password + Encrypted output

Screenshot 2 Symmetric Decryption:

Symmetric Decryption - Encrypted text + Password + Decrypted message

Screenshots 3 Asymmetric Encryption:

Asymmetric Encryption - Message + Public Key + Encrypted output

Screenshot 4 Asymmetric Decryption:

Asymmetric Decryption - Encrypted text + Private Key + Decrypted message

Q2 SCREENSHOT1 Symmetric Encryption (AES)

Text

Document / File

Encryption Type

Password (AES)
Symmetric · Password-based

Public Key (RSA)
Asymmetric · RSA-OAEP

PBKDF2 (100k iterations) + AES-256 · All processing in browser

AES Mode

AES-256-GCM
Authenticated (recommended)

AES-256-CBC
Classic block cipher

Plaintext

Audit findings are ready for review

Password / Passphrase

.....

Excellent

Encrypt

AES-256 · Ciphertext · Base64

Copy

AZayoXiVoYlSsXjBGx/DeuWqZqs7hYpYgvETmbkHoXR8hDw+17e8KCSX1yYqMgerSMrsR9dTafT14X/Ye9fhqAtybB96Du7VuuxKyLRDwB=

Contains: mode marker + salt (16B) + IV + ciphertext. Paste into the Decrypt tab to recover.

Security Notice

✓ All operations run 100% in your browser using the Web Crypto API

✓ No data is ever sent to any server — fully offline capable

✓ PBKDF2 with 100,000 iterations for key derivation

✓ AES-256-GCM provides authenticated encryption

Q2. SCREENSHOT 2- SYMMETRIC DECRYPTION

Hash Encrypt Decrypt Encode / Decode Key Generator

1 Decrypt

Input Type

Text

Encrypted File (.enc)

Decryption Type

Password (AES)
Symmetric · Same password

Private Key (RSA)
Asymmetric · RSA-OAEP

Auto-detects AES-GCM vs AES-CBC from the ciphertext header

Ciphertext (Base64)

AZayoXiVoYlSsXjBGx/DeuWqZqs7hYpYgvETmbkHoXR8hDw+17e8KCSX1yYqMgerSMrsR9dTafT14X/Ye9fhqAtybB96Du7VuuxKyLRDwB=

Password / Passphrase

.....

Decrypt

2 Output

Decrypted Plaintext

Copy

Audit findings are ready for review

Security Notice

✓ All operations run 100% in your browser using the Web Crypto API

✓ No data is ever sent to any server — fully offline capable

✓ PBKDF2 with 100,000 iterations for key derivation

✓ AES-256-GCM provides authenticated encryption

Q2. Screenshot 3 - Asymmetric Encryption (RSA with Public Key)

Input type

Text

Document / File

Encryption Type

Password (AES)
Symmetric · Password-based

Public Key (RSA)
Asymmetric · RSA-OAEP

RSA-OAEP with SHA-256 · Encrypt with public key, decrypt with private key

Plaintext

This is a secure message for the Senior IT Auditor

RSA can encrypt up to ~190 bytes (2048-bit key) or ~446 bytes (4096-bit key)

Public Key (PEM)

-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAkpo0lcs+1jLLskZH0A1
GdPP10Gg7VjcrZy7dHlFA0nz/k/pknPx007prR3n3zVFCV5g5QUPUvzLh4DBF
ubR1LVDHkVsQc9/7xV7La/y4Q0mPyTz7334oG0lYNg0bvv+iE1NT6iR2U06vJQ
EmmcoEi14QwQYyQR7W00wFqH3F33E2Vs+hrhpd1g8uq0YnIIN5kAcxY9M0b9V8
8oLID097BmtcFH0K20TuhEEmwskMF+DGF3I7XPZjQX12cNwLhuk71bCC31HDFLLS

Generate a key pair in the Key Generator tab → RSA Pair

Encrypt with Public Key

RSA-OAEP · Ciphertext · Base64

Copy

T/Asa1M8nB87bWle02pr3jvShIF1YsUPWbmyCOF-r71BXR8pdjwAh7rnqPHCuKmq1/Im0FD
KbEw1BD1zf9N8rSHSLSK9LIP0QzLU3qIYDLtTT2sY143XkMqNZ5dhhb+qRoT3APyGTGC0
s6teYd7FE70upw3Uq+X3E+t7q087L3COF/ISppbfMRvUwK4zspe1+cspnEyc8046Z3FC
cKF2vCKPRV/0MYK+owUCYH3Q02V5IpEZ8awq8Hv3zmP7KktA0Z+cCdNlKETgo889m571I
RNXUJftsymb8uCC1kxvOF7L76QjFttd21Eo+BrGQREpg8X16CjRQ1v+IwJ216Q==

RSA ciphertext. Decrypt using the corresponding private key in the Decrypt tab.

Security Notice

All operations run 100% in your browser using the Web Crypto API

No data is ever sent to any server — fully offline capable

PBKDF2 with 100,000 iterations for key derivation

AES-256-GCM provides authenticated encryption

Q2. Screenshot 4 Asymmetric Decryption (RSA with Private key)

1 Decrypt

Input Type

Text

Encrypted File (.enc)

Decryption Type

Password (AES)
Symmetric · Same password

Private Key (RSA)
Asymmetric · RSA-OAEP

Decrypt RSA-OAEP ciphertext using your private key

Ciphertext (Base64)

gQ8bQ/TtkGyNEK+0T6HJh8BkK4CR1boP0kLID09t5p6+1Xh+uNw/55GhAch/CuI7KQaoMY133N1a5ana/+0beRt08X/UH4CHe10zYem12Ts
4m7/QUF66aCkH9pT0okHtHhpg03FErIV61j12q1KJNF8htx0PPYn11x1Fha1v3ASvQKH396yY555DcKqfVFC1Lum5EqmQmrv9q1Yj+g
DYe4N+63Rnzj1Zq2B0W69T+mjh+gV8ctsdQjE2X8ju4hZ31p4pN9mDU6vYH59RnrFckctxe7bmnAQQLDtytA0De7tPEDbh/nxfHjZ0Kdwg
qOPW5ZdQ==

Private Key (PEM)

-----BEGIN PRIVATE KEY-----
MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBKcwggSjAgEAAoIBAQ7C3uRB2K3qa0t
EGgtzeblCH7jmeCYr3kzyeOCalevOyIoD1ATBqAMuyCkTVFUOTIS9KQlMg33mU
IM0Fudvubqmb3k08LYBq7Kta0sGEjfoX71gdPdePrGgJUEvcv32+aAvLMA26YU
QEXowCmFUSH+skTFgAZ0qj1Mz+cVcLLjYmQh3HqK02yhovS04MAsqUAc5Zp0
EULCcRc+7EdoFLD0tLKV8Y0ArCoDKvZf171GhHsZ0vv1v1n+an5s4darBMD08Rhu

Private key corresponding to the public key used for encryption

Decrypt with Private Key

2 Output

Decrypted Plaintext

Copy

This is a secure message for the Senior IT Auditor

Security Notice

All operations run 100% in your browser using the Web Crypto API

No data is ever sent to any server — fully offline capable

PBKDF2 with 100,000 iterations for key derivation

AES-256-GCM provides authenticated encryption

QUESTION 3

Audit Trail & Cryptographic Protection in AdVerse Sign

The AdVerse Sign platform provides a comprehensive audit trail that ensures the integrity, accountability, and security of the digital signing process. Below is an explanation of how each security principle is supported:

1. Accountability

How it's supported: The document shows the exact date and time each signer signed:

- Margaret Blay – Senior IT Auditor: **Friday, August 28, 2026**
- Staff IT Auditor: **Friday, August 28, 2026**

This creates a clear, undeniable record of **who signed** and **when they signed**, making each signer accountable for their action.

2. Traceability

- ***How it's supported:*** The platform provides complete traceability through:
 - **Creation date:** Thursday, August 27, 2026 (when the document was uploaded)
 - **Signer details:** Names, email addresses, and signing timestamps
 - **Verification Code:** TBQS30NS (proof that the signing process was authenticated)

You can trace the entire document lifecycle from creation to completion.

3. Integrity

How it's supported: The platform provides cryptographic hashes to verify document integrity:

Hash Type

Hash Value

Document Hash (SHA-256)	cd15f82d7549ac387f2c9d725811220749f24fc37f4b96032b662a57fc25d92c
Final Signed Hash (SHA-256)	968715ee629884dfbf2f27a562700b077befa81849a920577d3e26c0ee98fa34

If the document is altered after signing, these hashes will change, making the signatures invalid. This ensures the integrity of the document.

4. Monitoring

How it's supported: The platform allows administrators to monitor;

- Document status (SIGNED)
- Signer progress (both completed)
- Signing timestamps
- Cryptographic validation status

This allows real-time monitoring of the signing process.

5. Investigation

How it's supported: In case of disputes (e.g., a signer denies signing), the audit trail provides objective evidence:

Signer identities and emails

Signing timestamps

Cryptographic signature validation (Ed25519 Valid)

Verification codes

This makes investigations straightforward and reliable.

6. Audit/Compliance Evidence

How it's supported: The platform provides legally admissible evidence:

- **Ed25519 cryptographic protection** – a strong digital signature algorithm
- **SHA-256 hashing** – industry-standard for integrity verification
- **Complete signer records** with timestamps
- **Cryptographic Signature Validation** showing "Valid" for both signers

This meets compliance requirements for standards like:

- **eIDAS** (EU electronic signature regulations)
- **GDPR** (data protection)
- **SOX** (Sarbanes-Oxley Act)
- **HIPAA** (healthcare data protection)

Summary of Cryptographic Terms

Term	Meaning
SHA-256	A hash function that creates a unique "fingerprint" of the document. If the document changes, the hash changes.
Ed25519	A digital signature algorithm that provides strong cryptographic protection for electronic signatures.
Hash Integrity	Hash values ensure the document has not been tampered with after signing.

Conclusion

In conclusion, I successfully completed the digital signature process using the AdVerse Sign platform. The document (Audit_Report_2026) was uploaded, two signers were added (Margaret Blay – Senior IT Auditor and Staff IT Auditor), signature fields were placed, and both signers signed the document. The platform generated SHA-256 hashes to ensure document integrity and used Ed25519 cryptographic signatures for authentication. The audit trail provides accountability, traceability, integrity, monitoring, investigation,

and compliance evidence. The verification code for this document is **TBQS30NS**.

02:53

LTE

Gmail



services.getadverse.com



Audit_Report_2026

SIGNED

Created: Thursday, August 27, 2026

Integrity Hashes

Document Hash (SHA-256)

cd15f82d7549ac387f2c9d725811220749f24fc37f4b96032
b662a57fc25d92c

Final Signed Hash (SHA-256)

968715ee629884dfffb2f27a562700b077befa81849a920577
d3e26c0ee98fa34

Signers



Margaret Blay – Se...

M*****3@g...

Friday, August 28, 2026

signed



Staff IT Auditor

c*****7@gmail.c...

Friday, August 28, 2026

signed

Cryptographic Signature Validation



Margaret Blay – Senior IT Auditor

Ed25519 Valid



Staff IT Auditor

Ed25519 Valid



Powered by AdVerse Sign · SHA-256 + Ed25519 Verification



Ask



02:45

LTE



services.getadverse.com



Sign Document

corrielay7@gmail.com

Draw

Type

Upload

Previewing on the document — check the highlighted fields.

Corrie Adelaide Blay

Corrie Adelaide B...

Corrie Adelaide Blay

Corrie Adelaide ...

Corrie Adelaid...



I agree that my electronic signature is legally binding and I have reviewed the document. I understand this action cannot be undone.



Sign document

Powered by AdVerse Sign · Ed25519 cryptographic protection



Ask



02:52

LTE



Signed: Audit_Report_2026

Inbox



YESYOU CAN CYBERSEC...

02:45

to me



AdVerse Sign

Your signed document

Hello **Staff IT Auditor**,

The document **Audit_Report_2026** has been signed by all parties. The final signed copy is attached to this email, with the signing certificate as its last page.

Verification code

TBQS30NS

Anyone can confirm this document is authentic and unaltered at
<https://services.getadverse.com/verify-document/TBQS30NS>.

Reply

Forward



02:44

LTE



services.getadverse.com



Sign Document

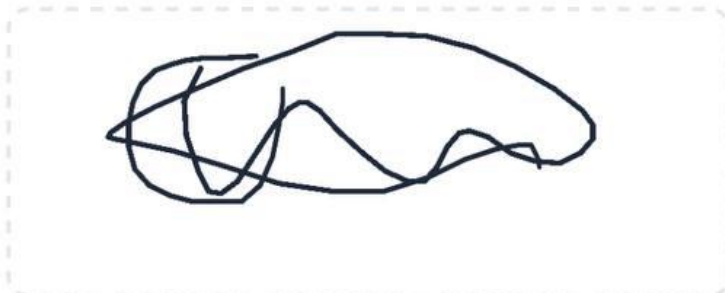
corrielay7@gmail.com

Draw

Type

Upload

Previewing on the document — check the highlighted fields.



Signature captured — previewing on the document

Clear

☒ I agree that my electronic signature is legally binding and I have reviewed the document. I understand this action cannot be undone.

Sign document



Powered by AdVerse Sign · Ed25519 cryptographic protection



Ask



02:43

LTE



services.getadverse.com



AdVerse Sign

Enter Verification Code

Check your email for the 6-digit code

7 8 6 4 0 1

Verify Code

Didn't get it? Codes can take a minute to arrive — check your spam folder. Each new code replaces the previous one.

Resend code in 10s

Powered by AdVerse Sign · Ed25519 cryptographic protection



Ask



9



Audit_Report_2026

✓ Reviewed



Page 1 / 1



100%



Margaret Bl...

Audit Report 2026

2026-0...

Prepared by: Staff IT Auditor

Reviewed by: Senior IT Auditor

Date: August 2026

Text

This report summarizes the findings of the annual audit.

Your initials

Your signature



Install AdVerse



To install this app on your iPhone/iPad:

- 1 Tap the  Share button
- 2 Scroll down and tap  "Add to Home Screen"



10



2:36

LTE

services.getadverse.com

1

Sign Document

Margaretblay3@gmail.com

 Draw

 Type

 Upload

 Previewing on the document — check the highlighted fields.



Signature captured — previewing on the document

Clear

☒ I agree that my electronic signature is legally binding and I have reviewed the document. I understand this action cannot be undone.

 Sign document



Powered by AdVerse Sign · Ed25519 cryptographic protection

02:35

◀ Gmail

LTE



services.getadverse.com



 AdVerse Sign

Verify Your Identity

Audit_Report_2026

A verification code will be sent to
corrieblay7@gmail.com

Send Verification Code

🔒 Powered by AdVerse Sign · Ed25519 cryptographic protection



🔍 Ask



9

02:45

LTE



services.getadverse.com



AdVerse Sign



Document Signed

Your signature has been applied with cryptographic protection.

Verification Code

TBQS30NS

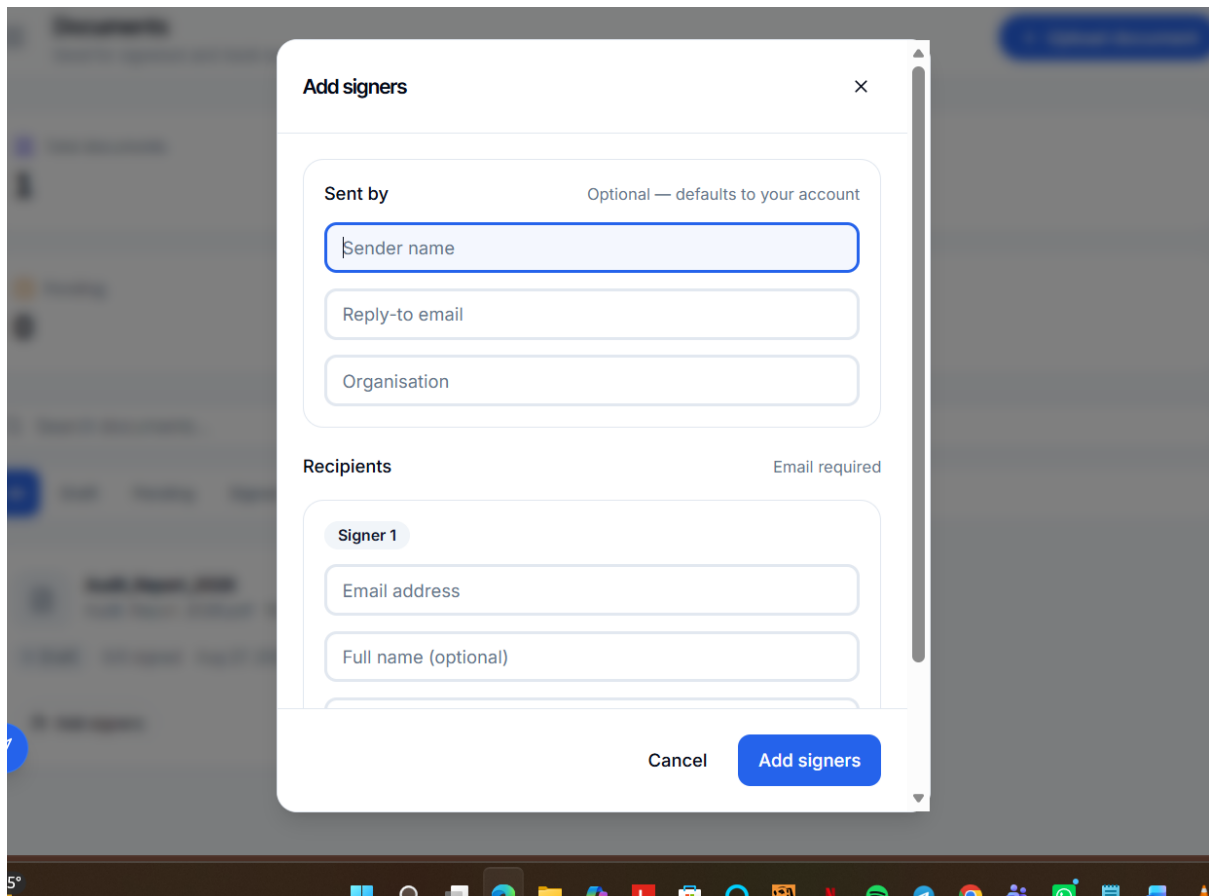
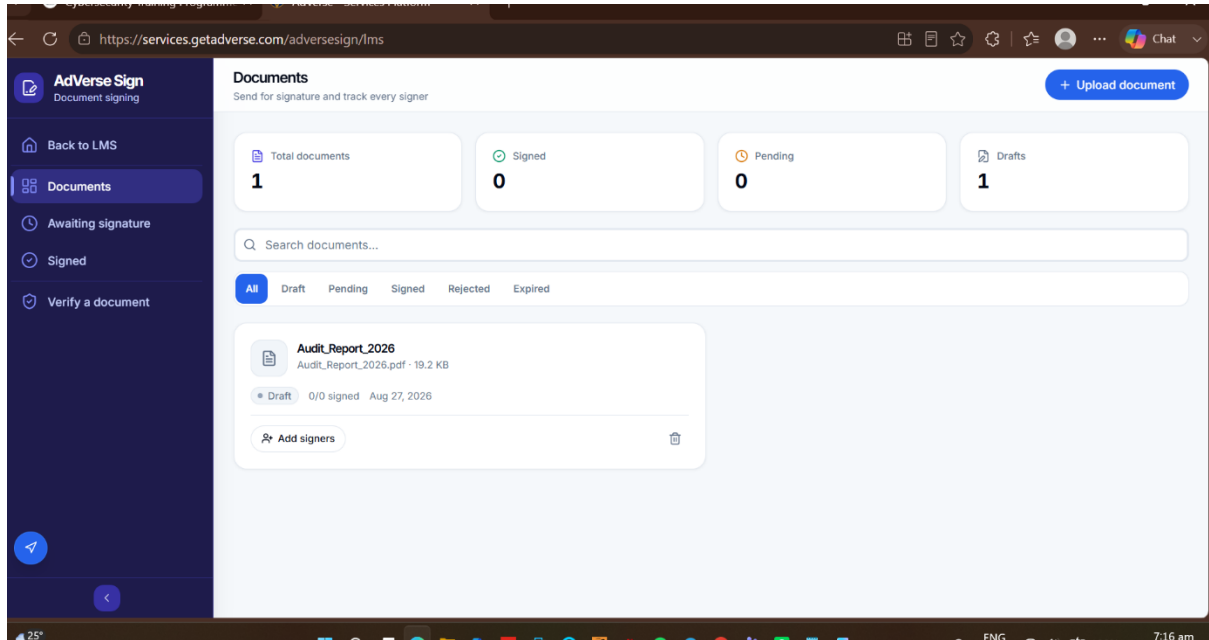
You may close this page. A copy will be available once all parties have signed.

Powered by AdVerse Sign · Ed25519 cryptographic protection



Ask





Add signers



Recipients

Email required

Signer 1

Remove

Margaretblay3@gmail.com

Margaret Blay – Senior IT Auditor

+233552251244

Signer 2

Remove

corrieblay7@gmail.com

Staff IT Auditor

Cancel

Add signers



AdVerse Sign
Document signing

Back to LMS

Documents

Awaiting signature

Signed

Verify a document

Documents

Send for signature and track every signer

+ Upload doc

Total documents
1

Signed
0

Pending
1

Drafts
0

Search documents...

All Draft Pending Signed Rejected Expired

Audit_Report_2026
Audit_Report_2026.pdf - 19.2 KB

Pending 0/2 signed Aug 27, 2026

Open >

25°

ENG US

100%

AdVerse Sign
Document signing

Back to LMS

Documents

Verify a document

Audit_Report_2026

Audit_Report_2026.pdf

Pending

Details Signers Fields Audit trail

M Margaret Blay - Senior IT Auditor Pending
Margaretblay3@gmail.com

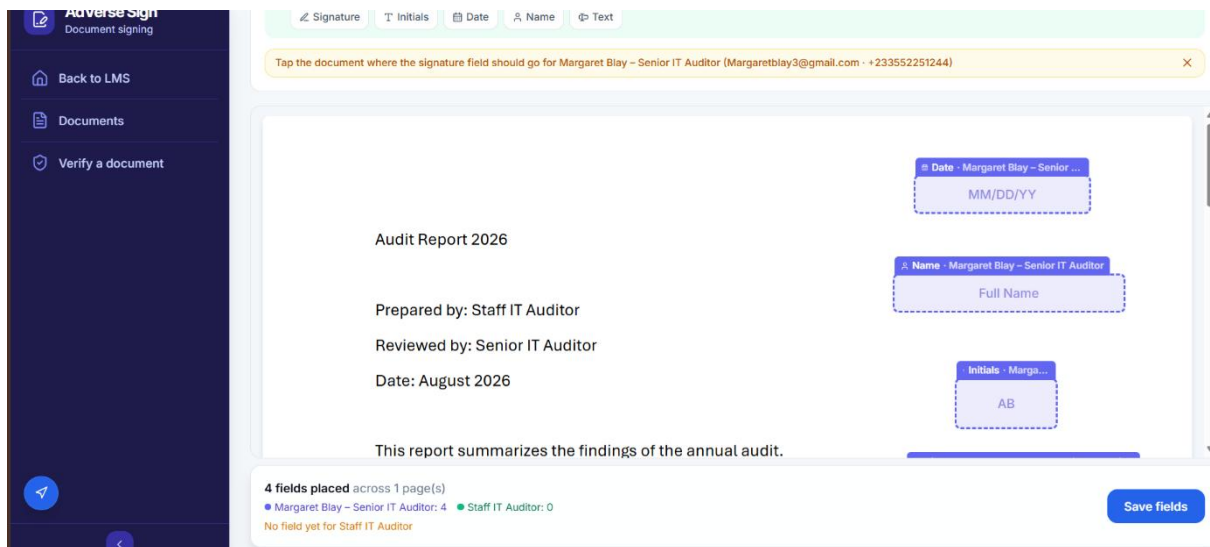
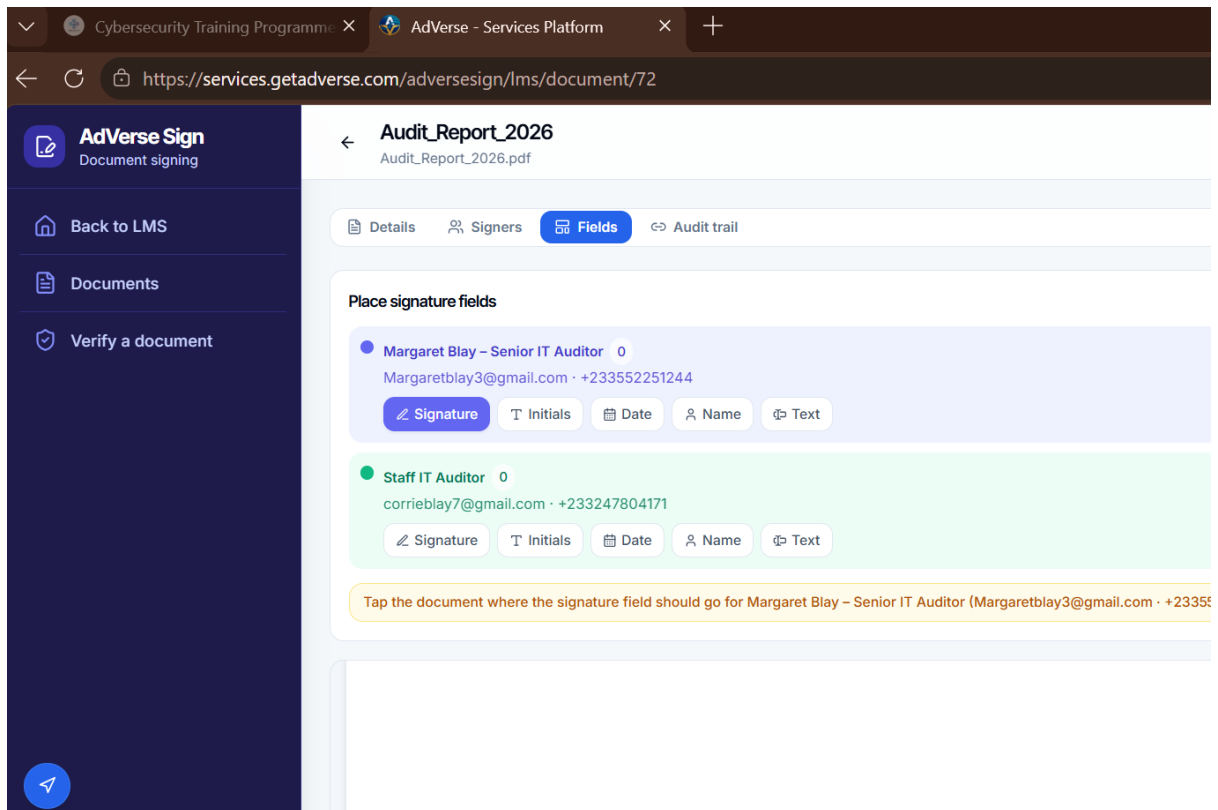
Order 1 Edit Send link Revoke

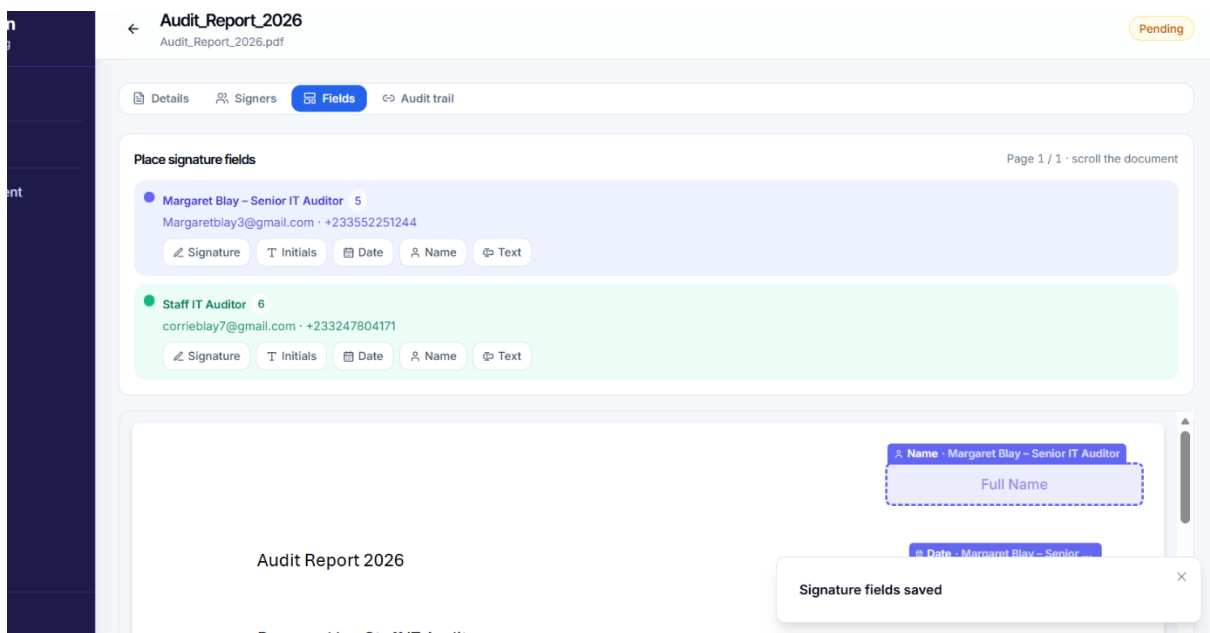
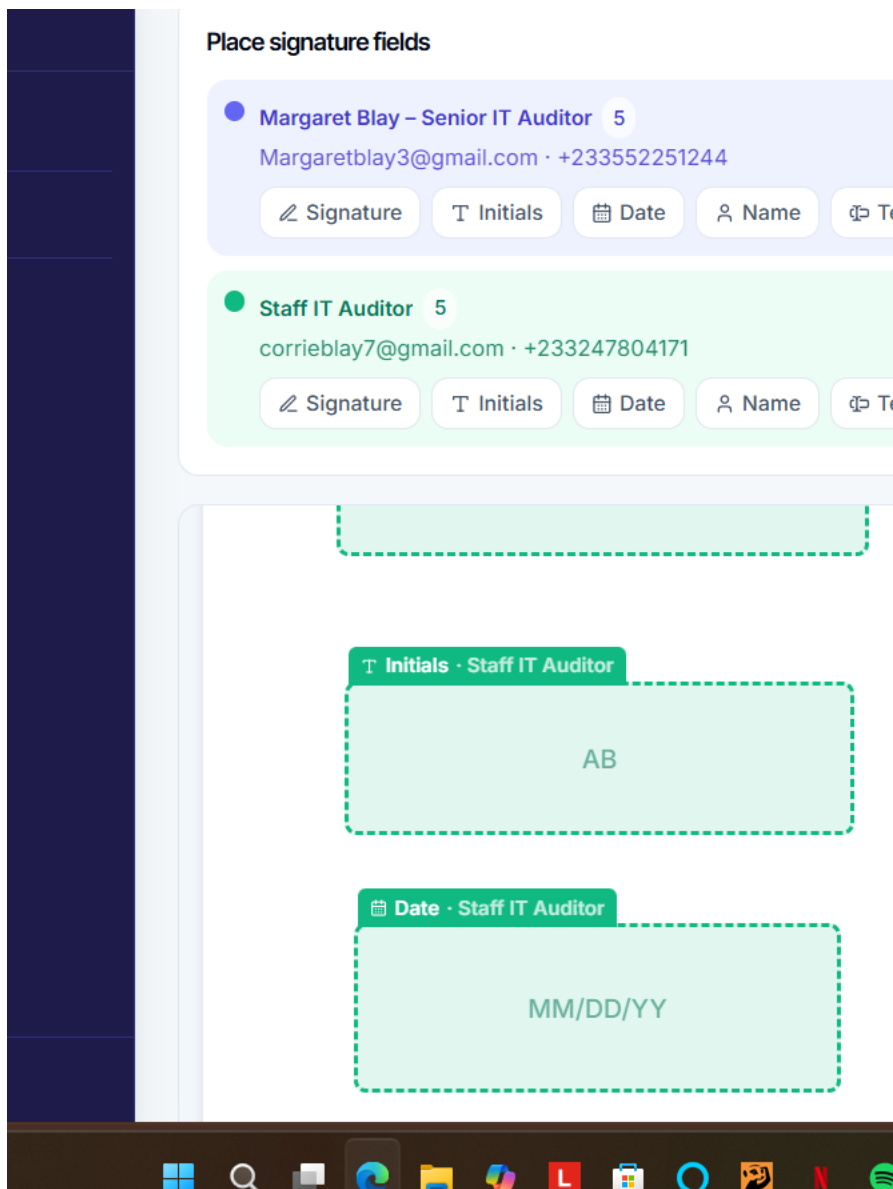
No signature field placed for this signer. Place one in the **Fields tab** so their signature lands where you want it — otherwise it is added at the bottom of the last page.

S Staff IT Auditor Pending
corrieblay7@gmail.com

Order 2 Edit Send link Revoke

No signature field placed for this signer. Place one in the **Fields tab** so their signature lands where you want it — otherwise it is added at the bottom of the last page.





Audit_Report_2026

Audit_Report_2026.pdf

Pending

Details

Signers

Fields

Audit trail

Place signature fields

Page 1 / 1 · scroll the document

Margaret Blay - Senior IT Auditor

5

Margaretblay3@gmail.com · +233552251244

Signature

Initials

Date

Name

Text

Staff IT Auditor

6

corrieblay7@gmail.com · +233247804171

Signature

Initials

Date

Name

Text

Audit Report 2026

Prepared by Staff IT Auditor

Name: Margaret Blay - Senior IT Auditor

Full Name

Date: Margaret Blay - Senior

Signature fields saved

AdVerse Sign

Document signing

Back to LMS

Documents

Verify a document

Audit_Report_2026

Audit_Report_2026.pdf

Pending

Details

Signers

Fields

Audit trail

M

Margaret Blay - Senior IT Auditor

Pending

Order 1

Edit

Send link

Revoke

S

Staff IT Auditor

Pending

Order 2

Edit

Send link

Revoke

 **AdVerse Sign**
Document signing

 Back to LMS

 Documents

 Verify a document



Document information

Title

Audit_Report_2026 

Verification code

TBQS30NS

Document hash

cd15f82d7549ac387f2c9d725811220749f24fc37f4b96032b662a57fc25d92c 

Created

Aug 27, 2026, 07:14 AM

File size

19.2 KB

Final signed hash

968715ee629884dfffb2f27a562700b077befa81849a920577d3e26c0ee98fa34

Actions

 **View PDF**
Preview the document as uploaded

 **Download signed document**
Your document with the signatures in place — nothing else added

 **Download certificate only**
Certificate of completion — identities, timestamps and the audit trail

 **Download both together**
Signed document followed by the certificate, as one file

 **Resend signed copy**
Email the signed PDF or SMS a secure download link to the signers

 **Duplicate document**
Make an editable copy with the same fields and signers

 **Save as template**
Reuse this layout for a new client from New → Use a template

Public verification code

TBQS30NS

 Copy

QUESTION 4

The first screenshot;

- The padlock icon  in the address bar shows;
- "Connection is secure" message
- "Certificate is valid" confirmation
- (Website name) FirstBank Ghana website

Part 1: Security Measures for Online Banking

Security Measures I Look for at FirstBank Ghana Online Banking

1. Padlock Icon & HTTPS When I log in, the first thing I check is the padlock symbol in the browser's address bar. It tells me the connection is secure. The website address starts with `https://`, which means any information I send—like my password or account details—is encrypted. If I click on the padlock, I can see that the certificate is valid, which reassures me that I'm really connected to FirstBank.

2. Login Credentials To get into my account, I need to enter my email and password

This is the first layer of protection. I like that the password box hides the characters with dots, so even if someone is nearby, they can't see what I'm typing.

3. Safe Practices I make sure to type the official FirstBank website address directly into my browser instead of clicking links from emails or text messages—those could be phishing attempts. I also use a strong, unique password that I don't reuse anywhere else. And if FirstBank offers Multi-Factor Authentication (MFA), I'll turn it on, because it adds an extra step (like a code sent to my

phone) that makes it much harder for anyone else to break into my account.

Part 2 - How Browser Interacts with Website

When I type a web address into my browser, here's what happens behind the scenes:

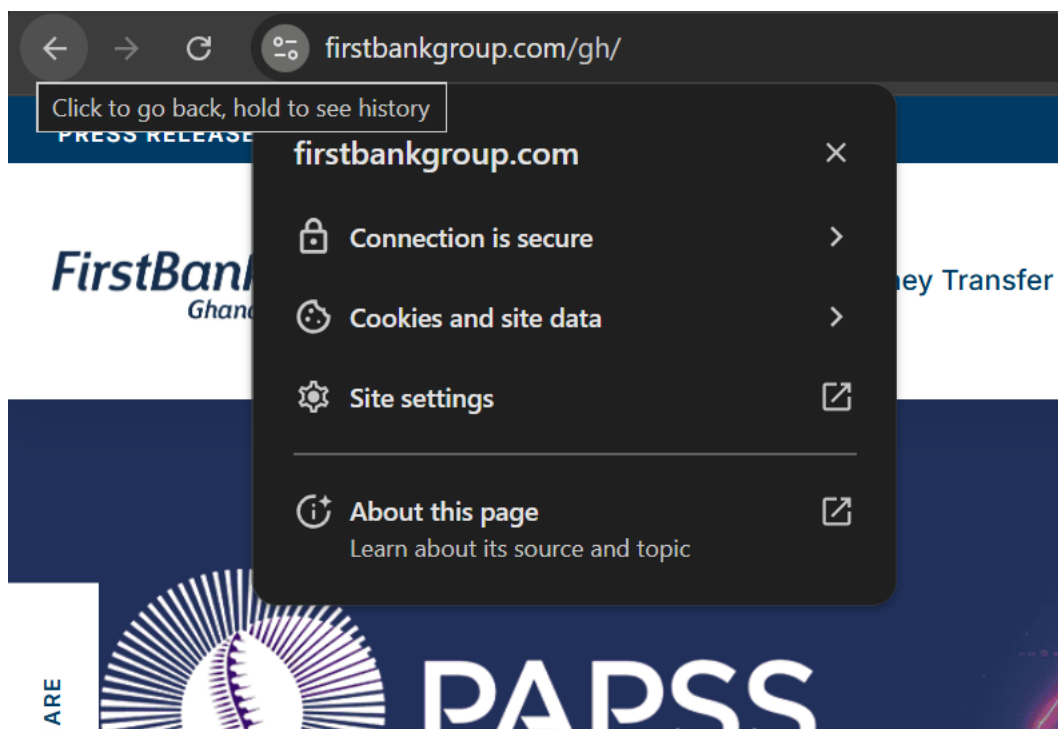
- 1. Finding the Server (DNS Lookup)** The browser first figures out where the website lives. It takes the name I typed (like `www.firstbank.com.gh`) and looks it up in the Domain Name System (DNS) to get the actual IP address of the server.
- 2. Sending a Request (HTTP Request)** Once it knows the server's address, the browser sends a message asking for the page. For example, it might say: "GET /login".
- 3. Getting a Response (Server Reply)** The server receives that request and sends back the files that make up the page—things like HTML, CSS, JavaScript, and images.
- 4. Reading the HTML** The browser starts by reading the HTML file, which is basically the skeleton or structure of the page.
- 5. Loading Extra Files**, The HTML usually points to other files it needs—CSS for styling, JavaScript for interactivity, and images. The browser sends separate requests to grab each of those.
- 6. Building the Page (Rendering)** The browser puts everything together: HTML for structure, CSS for design and JavaScript to make things interactive. That's when the page appears on my screen.

7. Caching for Speed To make things faster next time, the browser saves some of those files locally. So, if I revisit the site, it doesn't have to download everything again.

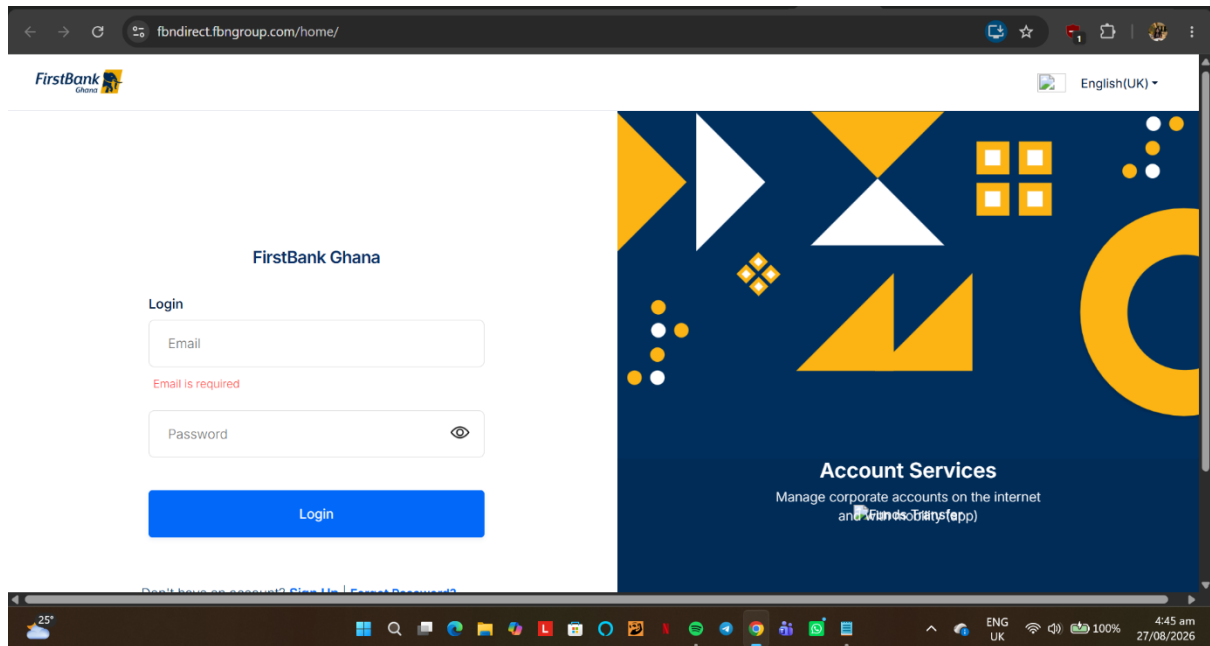
And if I open the Network tab in Developer Tools, I can watch all these requests happening in real time—seeing each file load as I refresh the page.

The browser uses **TCP/IP** to establish a connection with the server before sending the HTTP request.

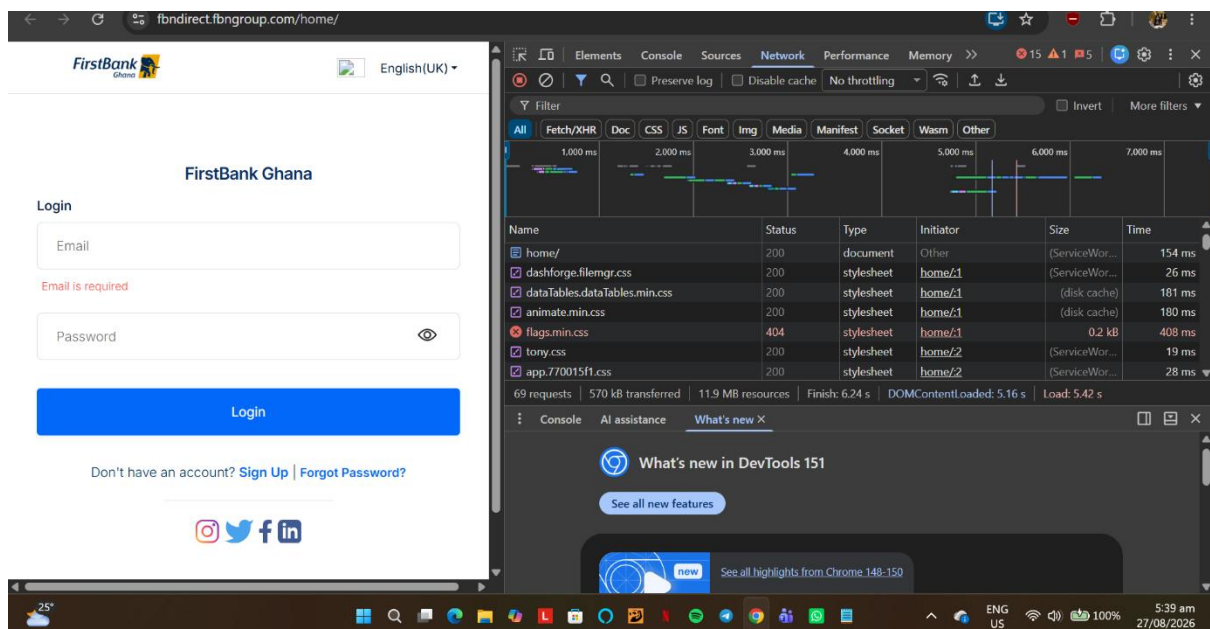
Q4 Screenshot1 Padlock and HTTPS Security



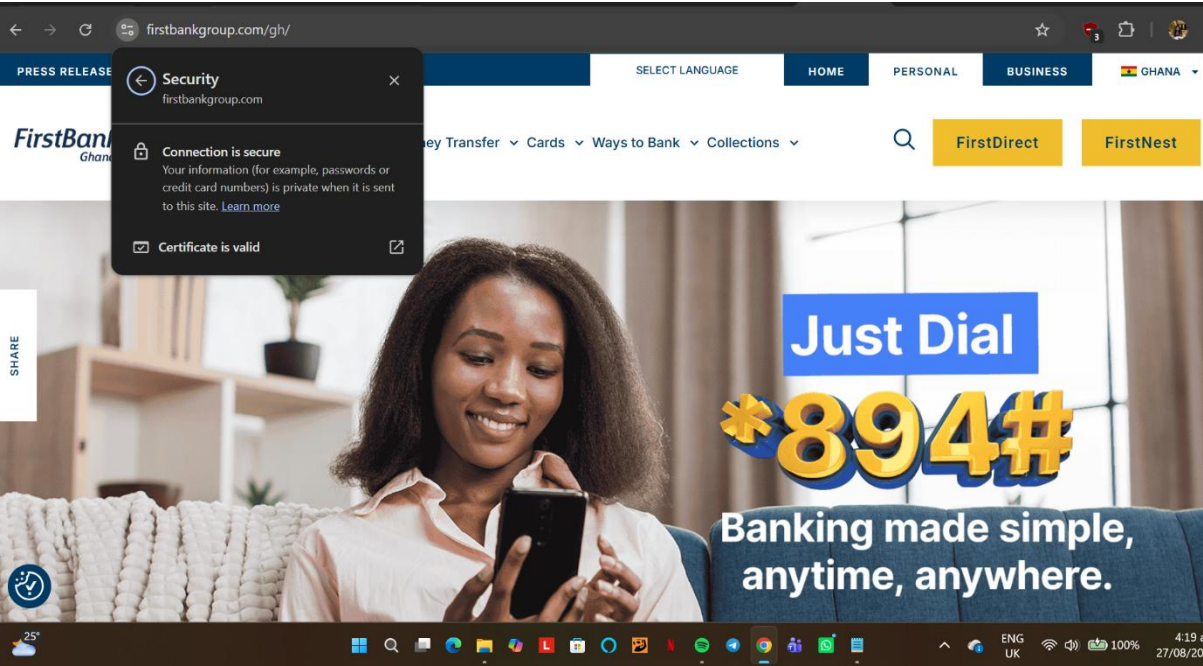
Q4 Screenshot 2 - Login Page Security



Q4 Screenshot 3 - Network Tab Browser Interaction



Q4 Screenshot 4



Question 5:

How WhatsApp Secures Communication

How WhatsApp Keeps My Chats Secure

WhatsApp protects everything I send; being it messages, calls, photos, and videos—using **end-to-end encryption**. That means my messages are scrambled on my phone before they leave, and only the person I'm chatting with can unscramble them on their phone. No one in between can read them—not hackers, not WhatsApp, not even Meta.

This encryption is built on the **Signal Protocol**, which is considered one of the most secure systems out there. One of the things I like about it is **Perfect Forward Secrecy**: every single message gets its own unique key. So even if someone somehow got hold of one key, it wouldn't help them read past or future messages.

The keys themselves are created and stored only on my device. They're not kept on WhatsApp's servers, which means WhatsApp doesn't have access to my conversations—it just delivers them in encrypted form.

Of course, encryption only protects my messages while they're traveling. Once they're decrypted on my phone, they're visible there. That's why it's still important to lock my phone with a strong password or use biometrics, so nobody can just pick up my device and read my chats.

WhatsApp uses a **hybrid model** — asymmetric encryption (via the X3DH key exchange) to establish the initial session, then

symmetric encryption for the actual message content via the Double Ratchet.

These are the key point that should be covered

- a. End-to-End Encryption
- b. Signal Protocol
- c. Perfect Forward Secrecy
- d. Keys stored on my device
- e. WhatsApp can't read my messages